

Predicting Spotify's Song Popularity



This project aims to identify the key factors that influence a song's popularity on Spotify specifically, to predict a song's popularity.

By using [Spotify Songs Dataset](#), which contains data extracted from the Spotify API, we trained a machine learning model to predict a track's popularity based on its audio and metadata features

A machine learning model that can accurately predict the popularity score (regression task), ideally minimizing prediction error (e.g., RMSE or MAE) while also offering insight into what makes a song more likely to be popular is used.

This machine learning system developed in this project is designed to **predict the popularity score of a song** based on its audio features and metadata. It will connect to the Spotify API and will be able to predict whether a song has the potential to become popular on a scale of 1-100. It can provide value to several stakeholders in the music industry:

Artists and Music Producers Get early feedback on new tracks during the production phase which Helps make data-driven decisions about which songs to prioritize or promote.

Record Labels and Marketers Use Case: Evaluate a large catalogue of unreleased tracks to identify potential hits and Allocate marketing resources efficiently toward high-potential songs.

Streaming Platforms (e.g., Spotify) Use Case: Enhance recommendation systems or curate dynamic playlists based on predicted popularity by that Improve user engagement and satisfaction.

This document presents a comprehensive overview of the project, outlining each phase and emphasizing key findings:

- **Data Preparation:** Describes the preprocessing techniques applied to the original Kaggle dataset.
- **Exploratory Data Analysis (EDA):** Investigates patterns and relationships within the data through visualizations and statistical analyses.
- **Data Cleaning:** Covers the strategies used to handle missing values and remove outliers, ensuring data integrity.
- **Feature Engineering:** Details the development of new features aimed at boosting model performance.
- **Feature Selection:** Explores the application of regularization and feature importance methods to reduce overfitting and improve generalization.

Data preparation:

A “flat file” was prepared for further analysis and model training. We started with 32,833 records and 23 columns.

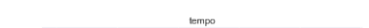
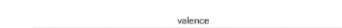
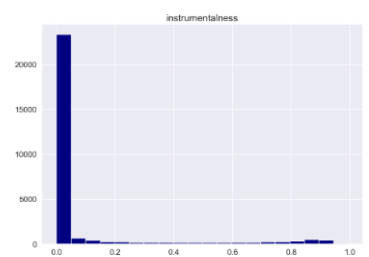
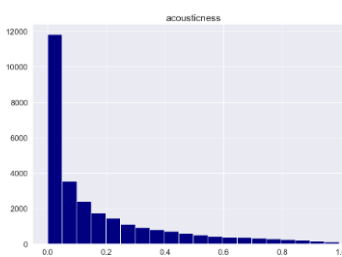
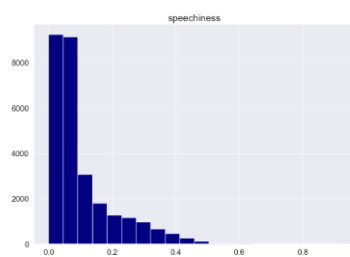
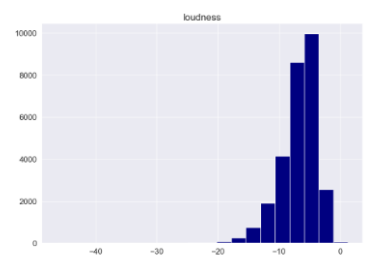
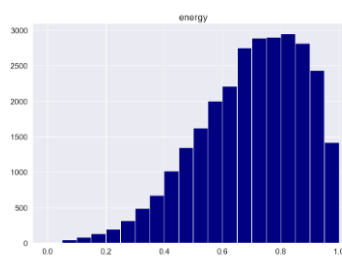
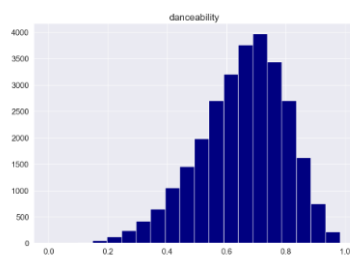
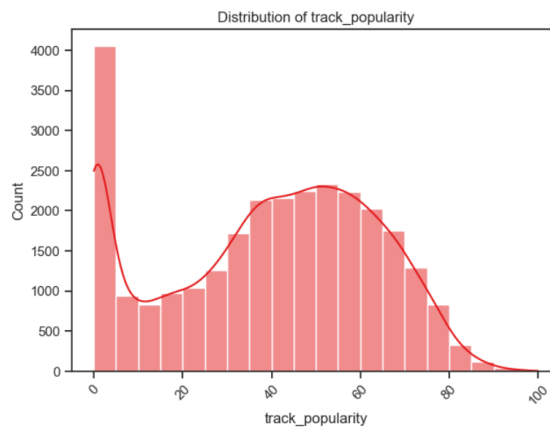
Key steps included:

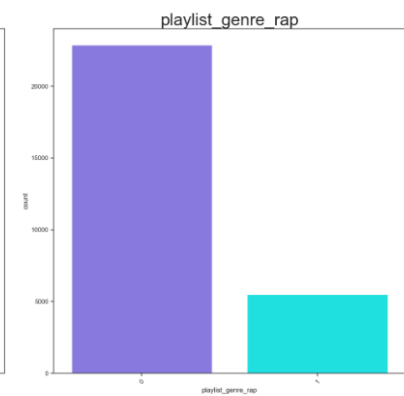
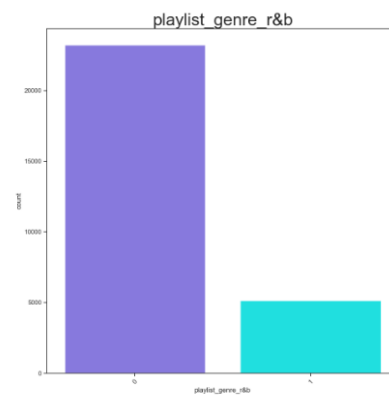
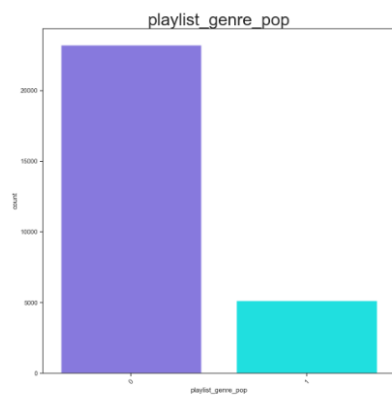
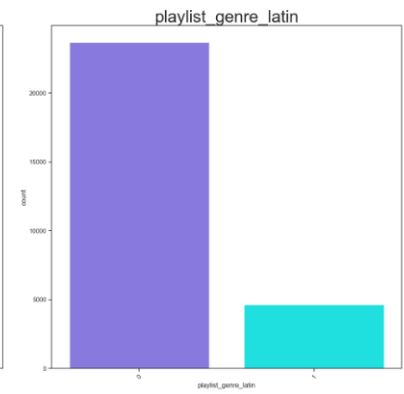
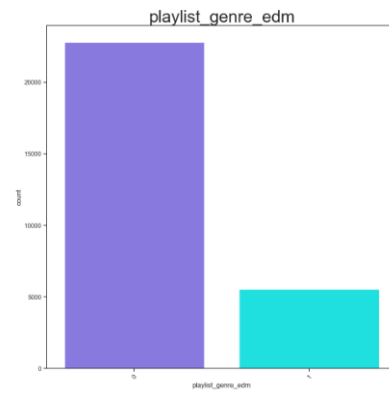
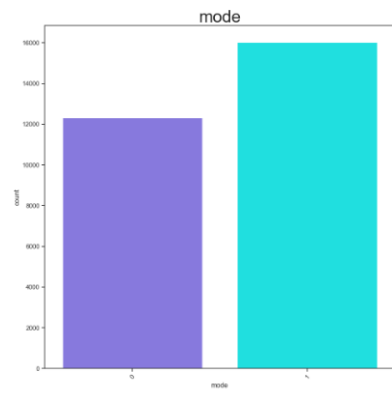
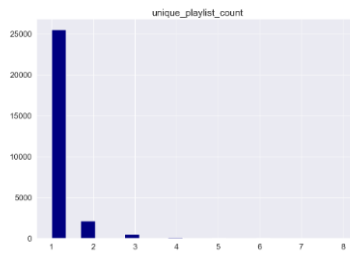
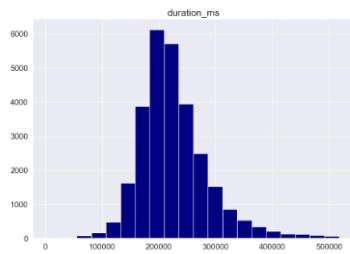
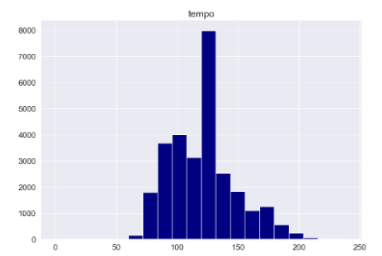
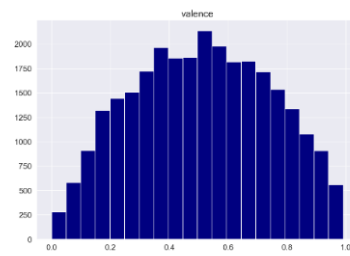
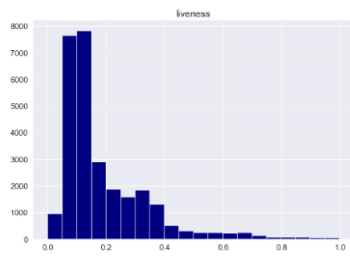
- Removing Bias contributing column “subgenre”.
- Creating a new ‘unique_playlist_count’ column that sums the number of playlists a song appears in to replace the duplicates of a single “track_id” found in multiple playlists.
- creating one hot encoding for “playlist genre” in order to aggregate “track_id” into a single row.
- normalize text by lowercasing, removes stop words, punctuation, and special characters, but keeps numbers and parentheses.
- Text based columns were extracted into a separate Data Frame to facilitate further feature engineering.

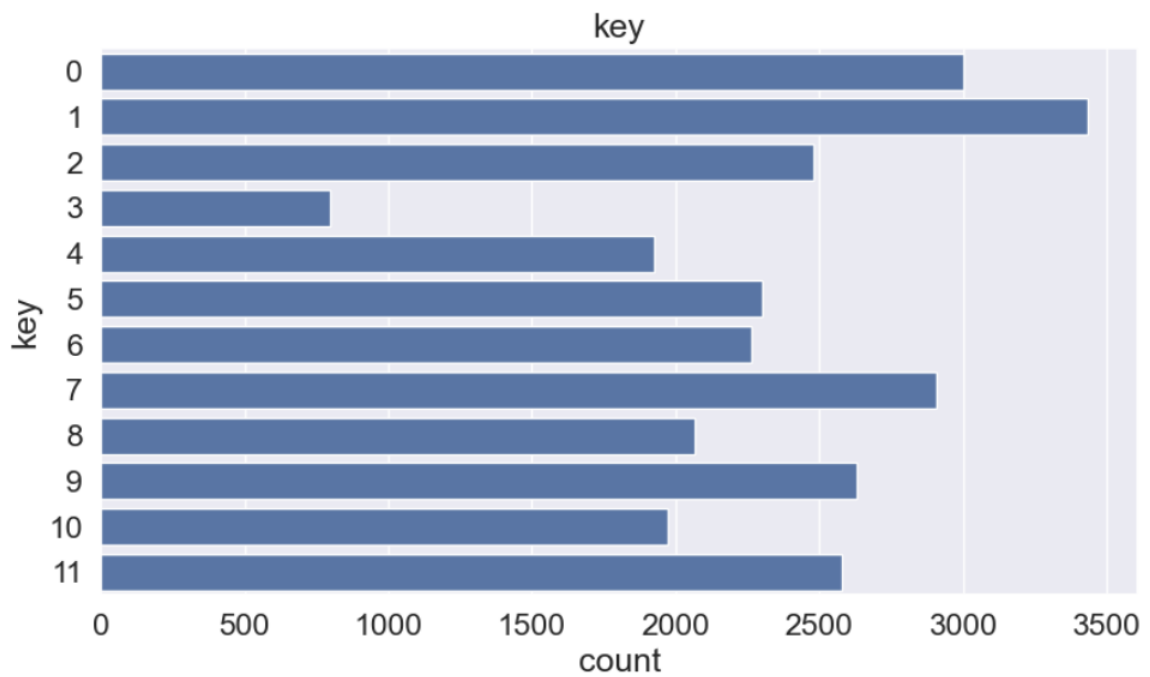
EDA – Explanatory Data Analysis

The dataset was explored to understand key distributions, correlations, and relationships between features. Main steps included:

- **General inspection** of feature types, unique values, and basic statistics.
- **Visualization of numerical and categorical distributions** using histograms and boxplots to detect skewness and outliers:

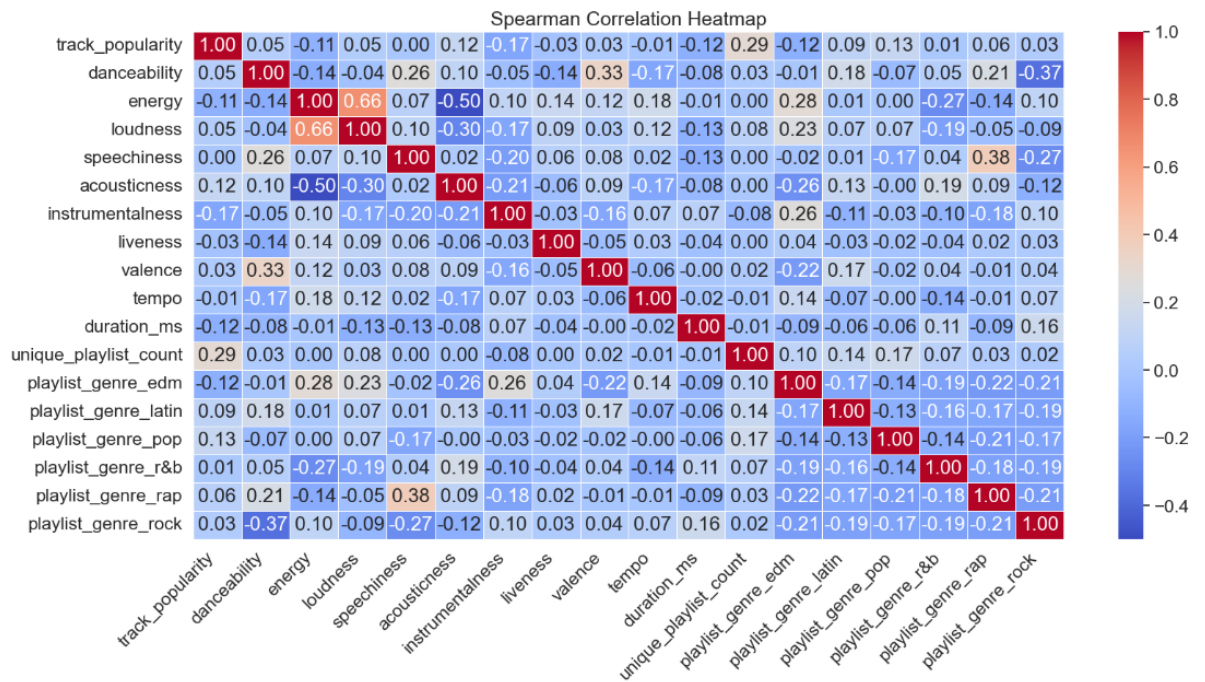






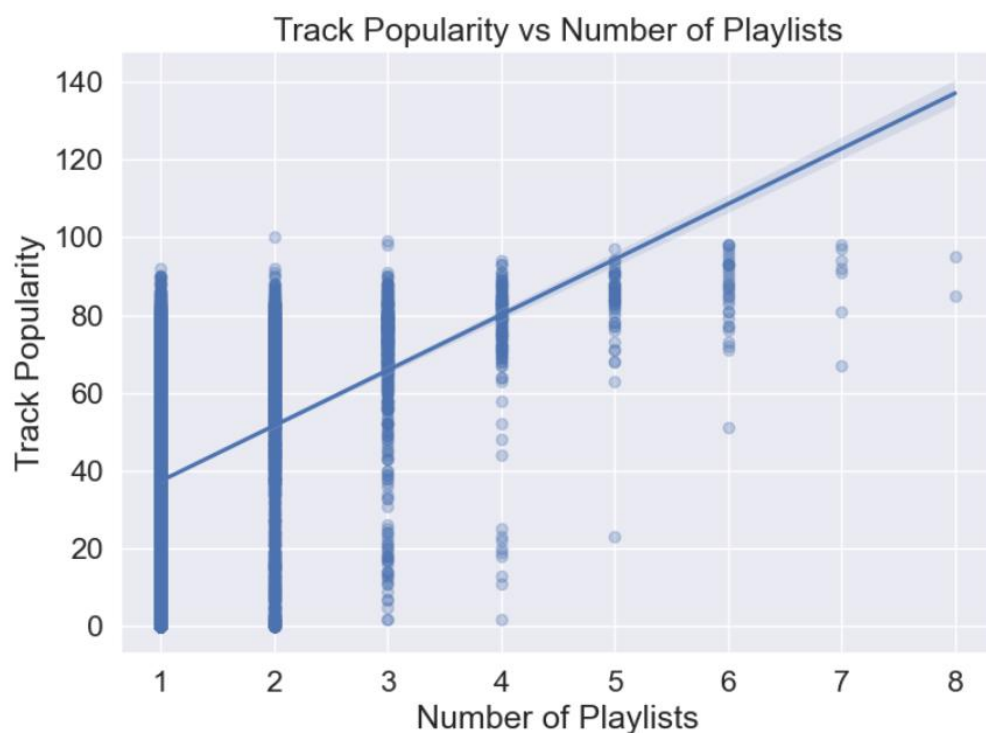
	skewness
unique_playlist_count	5.035169
instrumentalness	2.624985
liveness	2.081481
speechiness	1.964975
acousticness	1.576720
duration_ms	1.115164
tempo	0.513949
valence	-0.007334
danceability	-0.505789
energy	-0.645440
loudness	-1.359303

- **Correlation analysis** using a heatmap to identify relationships between audio features.



2008 was the year Spotify streaming service officially launched to the public. TTEST analysis revealed that the difference in track_popularity before and after 2008 is statistically significant.

The number of playlists a track is in, influences its popularity — positively and significantly.



Data Cleansing – Outliers and Missing Values

By visualizing popularity extremes across features (IQR method), this stage provided essential insights into feature behaviour and their potential impact on predicting track popularity. Outliers were detected and replaced by MICE algorithm for handling missing values.

Feature engineering

In this stage, the dataset was transformed to creatively extract more valuable data from the dataset and support proper modelling. Key steps included:

- Adding a new column, 'five popular words' which contains the five most frequently occurring words from the 'track name' column across all rows.
- scaling time values from milliseconds to seconds and extracting year and month from date.
- Adding a new column "before_2008" - Since the TTEST in the EDA showed a significant difference between songs that were released before 2008 and after.
- Adding a new column "number_tracks_artist" – this can give hints about the artist's fame or visibility. If an artist has lots of tracks in the dataset, they're probably more well-known, more active, or more likely to be picked up by Spotify's algorithm. Famous artists often release more songs, get more playlist placements, and have a bigger fanbase leading to higher popularity scores.
- Adding new column "release_year" - computes the number of years since each song was released. It will capture a different angle: Instead of absolute time, it captures how old a song is — which is often more relevant to popularity modeling.

Model Selection and finetuning

Identifying and prioritizing the most relevant features was performed by applying techniques like Lasso and Ridge regression, along with additional modeling approaches, we systematically selected a subset of 27 features to enhance overall model performance.

Model

This part contains the implementation, comparison, and fine-tuning of multiple regression models to predict Spotify song popularity using engineered features. several regression models were trained to predict track popularity, including Linear Regression, Decision Tree, Random Forest, AdaBoost, Gradient Boosting,

SVM, and XGBoost. The models were evaluated using four key metrics: MSE (Mean Squared Error), RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), and R^2 (R squared).

The results indicated that Random Forest outperformed all other models, achieving the lowest MAE (16.5), making it the most accurate model overall.

RandomizedSearchCV was employed to search for best hyperparameters.

There was a small Improvement on Finetuning (16.42). We will continue with the finetuned model. It should be mentioned The R^2 value was low (0.28) meaning the model explains small portion of the variance in the target. There's likely room for improvement by Feature engineering and/or Hyperparameter tuning.

Summary

This marks the conclusion of our machine learning journey.

We started by constructing a flat dataset, followed by exploratory data analysis to uncover meaningful patterns and insights. We then addressed data quality issues by handling outliers and missing values, and enhanced the dataset through thoughtful feature engineering. Using robust feature selection methods, we minimized overfitting and retained only the most impactful features. In the final stage, we selected and fine-tuned the best-performing model to deliver optimal predictive accuracy.